

1. Collision in Hashing

Definition

A **collision in hashing** occurs when **two different inputs produce the same hash output**.

- Hash functions take any input (like a file, password, or message) and produce a fixed-size output called a **hash** or **digest**.
- Example: A hash function that outputs 8-bit hashes can only produce 256 possible hashes, so two different inputs might produce the same hash.

Explanation

- Ideally, each unique input should produce a unique hash.
 - If two different inputs produce the same hash, that's called a **collision**.
 - Collisions can lead to security issues, especially in **digital signatures, password storage, and file integrity checks**.
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2. Collision Resistance

Definition

A hash function is **collision-resistant** if it is **computationally infeasible to find two distinct inputs that produce the same hash**.

Formally:

Hard to find $M_1 \neq M_2$ such that $H(M_1) = H(M_2)$ \text{Hard to find } M_1 \neq M_2 \text{ such that } H(M_1) = H(M_2)

Explanation

- Modern cryptographic hash functions (like SHA-256, SHA-3) are designed to be collision-resistant.
- Collision resistance is crucial for:
 - **Digital signatures** (so someone cannot forge a signed message)

- **Integrity checks** (so files cannot be tampered with unnoticed)
- **Password hashing** (so different passwords don't produce same hash)

3. Example of Collision in Hashing

Let's use a **toy hash function** to illustrate a collision:

Toy Hash Function

$H(M) = \text{sum of ASCII values of characters mod } 10$

Inputs

$M1 = \text{"Hi"} \rightarrow \text{ASCII sum} = 72 + 105 = 177 \rightarrow 177 \bmod 10 = 7$
 $M2 = \text{"AB"} \rightarrow \text{ASCII sum} = 65 + 66 = 131 \rightarrow 131 \bmod 10 = 1$
 $M3 = \text{"Ba"} \rightarrow \text{ASCII sum} = 66 + 97 = 163 \rightarrow 163 \bmod 10 = 3$
 $M4 = \text{"Gg"} \rightarrow \text{ASCII sum} = 71 + 103 = 174 \rightarrow 174 \bmod 10 = 4$

$\text{"AB"} \rightarrow \text{ASCII sum} = 65 + 66 = 131 \rightarrow 131 \bmod 10 = 1$
 $M3 = \text{"Ba"} \rightarrow \text{ASCII sum} = 66 + 97 = 163 \rightarrow 163 \bmod 10 = 3$
 $M4 = \text{"Gg"} \rightarrow \text{ASCII sum} = 71 + 103 = 174 \rightarrow 174 \bmod 10 = 4$

Now consider:

$M5 = \text{"Aa"} \rightarrow 65 + 97 = 162 \rightarrow 162 \bmod 10 = 2$
 $M6 = \text{"BB"} \rightarrow 66 + 66 = 132 \rightarrow 132 \bmod 10 = 2$  **Collision!**

- **Observation:** M5 and M6 are **different inputs** but produce the **same hash value (2)**.
- In a real-world scenario, this is why we need **strong collision-resistant hash functions** like SHA-256, where finding collisions is practically impossible.